Tangency closure and exact support representation checkpoint

Date: 2026-10-17T03:13:49.313081+00:00

Hoisted tangency completed all262144 source coordinates across all four fixed h,k profiles in 25.975039594999544 diagnostic cold seconds. No relaxed tangency candidate survived. Together with the previous exact Q-root closure, this excludes all repeated all-support norm quintics at those four profiles only. Squarefree and other outer cells remain open.

Search denominator inversions: 392866 versus 1307927 implied by the v1 formulas on the same complete prefix. These are scoped primitive counts, not whole-pipeline speed ratios; v1 never completed a matched cold reference. No A/A or qualified isolation was established.

Exact CRT representation uses sufficiently high total modulus degree. Its independent oracle passed27648 assignments and retains an insufficient-degree counterexample. The strong reference exploits real low-degree bounds rather than padded field arithmetic. Circuit profiles omit common norm generation, solving, extraction and original output costs from both arms.25percent clause-cut prediction supported: False.

n=83,d=18: direct clauses=44926,CRT clauses=124786,CRT/direct=2.777589814361394.

n=163,d=40: direct clauses=223376,CRT clauses=515382,CRT/direct=2.3072398109017977.

n=233,d=56: direct clauses=437792,CRT clauses=1034828,CRT/direct=2.3637435129011037.

n=283,d=70: direct clauses=684986,CRT clauses=1602140,CRT/direct=2.338938314067733.

n=409,d=100: direct clauses=1397828,CRT clauses=3218562,CRT/direct=2.302545091384634.

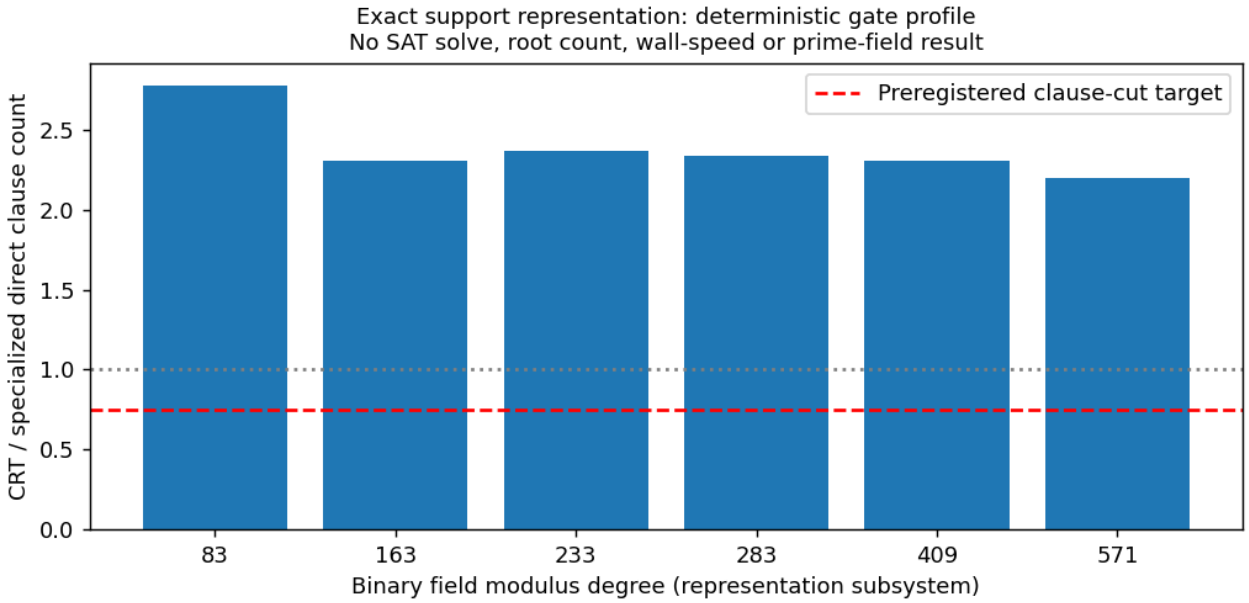
n=571,d=140: direct clauses=2741980,CRT clauses=6026060,CRT/direct=2.197703849043392.

Odd-characteristic root adapter:3624 independent small quintic checks passed with Hasse multiplicities, including inseparable and nonrational cases. Uses exact gcd(H,X^q-X), capped classical odd-field splitting, factor-product coverage and interval filtering. It returns unresolved on exhausted split attempts. NIST-width experiments are not run; source control CLI only. This is classical factoring, not a novel generalized solver.

Source and passing new integration tests are in PR2049; CI remains required. Novelty and qualified8x all-roots F4/F5/F6 superiority remain unestablished. Field-width circuit counts and four conditional norm profiles do not cover every curve/scale.

Next: native wide-prime/odd-extension root validation, complementary squarefree/global norm search, and fully matched hard baselines. Never multiply stage gains, use timeouts as ratios or discard refuted variants.

Reports remain local: automatic approval review rejected external report publication without explicit payload authorization. No alternate-format upload bypass is attempted.


Exact support representation: deterministic gate profile
No SAT solve, root count, wall-speed or prime-field result
Preregistered clause-cut target
CRT / specialized direct clause count
0.0
0.5
1.0
1.5
2.0
2.5
83
163
233
283
409
571
Binary field modulus degree (representation subsystem)